

ALI NIKKHAH

Senior Data Engineer | AI / ML Software Engineer

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PROFESSIONAL SUMMARY

Senior Data and AI Engineer with 5+ years of experience architecting large-scale data pipelines, Retrieval-Augmented Generation (RAG) systems, and ML platforms in production. Delivered predictive pipelines processing 10M+ events/day at Digital Turquoise and hybrid retrieval systems serving 5M+ daily queries at Digikala, lifting engagement by 7% and cutting processing latency by 40%. Trinity College Dublin research affiliate and Sharif University graduate, fluent in English (IELTS 8.0).

CORE TECHNICAL SKILLS

Languages: Python, Go, TypeScript, C++, SQL, Bash

Data Engineering: Apache Spark, Kafka, Airflow, dbt, ETL/ELT, Data Lakes, PeerDB

Databases & Warehousing: PostgreSQL, MySQL, MongoDB, Redis, BigQuery, Snowflake

ML / Deep Learning: PyTorch, TensorFlow, JAX, HuggingFace, vLLM, LangChain, LangGraph

RAG & Agentic AI: FAISS, ChromaDB, Pinecone, LlamaIndex, CrewAI, AutoGen, Multi-Agent Orchestration

MLOps & DevOps: Docker, Kubernetes, Terraform, MLflow, GitHub Actions, ArgoCD

Serving & Inference: FastAPI, Triton, TensorRT, ONNX Runtime, vLLM

Observability: Prometheus, Grafana, ELK Stack, Weights & Biases

PROFESSIONAL EXPERIENCE

Senior Data Engineer • Digital Turquoise

Tehran, Iran | Aug 2025 – Present

- Architected end-to-end predictive analytics pipelines processing 10M+ events/day, enabling real-time personalization across multimodal data sources.
- Engineered multimodal recommendation engines that lifted user engagement by 7% and improved click-through performance across product surfaces.
- Delivered scalable inference platform with drift monitoring, automated model versioning, and rollback workflows, improving model reliability and governance.
- Built real-time feature and embedding stores backing low-latency personalization for production traffic.
- Deployed data warehouses, Airflow DAGs, and PeerDB-based replication layers powering analytics and BI reporting.

AI Software Engineer → Senior AI Software Engineer • Digikala

Tehran, Iran | Jan 2024 – Aug 2025

- Architected scalable backend systems for hybrid retrieval, serving 5M+ daily queries with sub-second latency across e-commerce search.
- Pioneered Agentic Retrieval-Augmented Generation (RAG) frameworks combining FAISS vector search with keyword retrieval, boosting precision and recall.
- Engineered multi-agent orchestration coordinating retrieval, reasoning, and response-generation agents across evolving knowledge sources.
- Optimized data processing throughput by 40% via asynchronous batching, hierarchical caching, and continuous evaluation feedback loops.
- Implemented fault-tolerant orchestration for planning, retrieval, and verification stages with full traceability.

DevOps / Data Engineer • HomaCloud

Tehran, Iran | May 2021 – Feb 2022

- Built distributed MLOps pipelines on Kubernetes using Ray and MLflow for orchestration and experiment tracking.
- Automated CI/CD workflows with Terraform and ArgoCD, reducing deployment time to 10 minutes per release.
- Delivered scalable data-serving systems with A/B testing and phased rollout strategies for production safety.
- Implemented data versioning, feature stores, and end-to-end data-lineage tracking across the platform.

Backend Engineer • Mapna-MD2

Tehran, Iran | Nov 2020 – May 2021

- Developed RESTful APIs for industrial-IoT data ingestion, processing, and analytics across plant systems.
- Designed real-time dashboards and telemetry visualizations supporting operational decision-making.
- Improved query performance and reliability through optimized data handling and hardened auth flows.

RESEARCH EXPERIENCE

Machine Learning Researcher — EmoDub Project • Trinity College Dublin

Dublin, Ireland | May 2025 – Nov 2025

- Engineered transformer-based speech emotion recognition modules (wav2vec2.0, Whisper) for emotion-aware voice translation across languages.
- Architected multimodal fusion pipelines combining acoustic, semantic, and contextual embeddings for expressive cross-lingual dubbing.

Vision-Language Research Assistant • University of British Columbia

Remote (Tehran) | Feb 2024 – Nov 2025

- Built vision-language pipelines (ViT, BLIP, T5) for automatic ultrasound report generation aligned with DICOM standards.
- Achieved state-of-the-art BLEU and ROUGE-L scores; enhanced interpretability through attention visualization.

EDUCATION

B.Sc., Electrical Engineering — Digital Systems • Sharif University of Technology • Sep 2020 – Jul 2024

Overall GPA 3.65 | Major GPA 3.91

CERTIFICATIONS & LANGUAGES

- DeepLearning.AI: MLOps Specialization, Generative AI for Engineers, LLM Engineering.
- IBM Data Engineering Professional Certificate; Microsoft Data Warehouse.
- PostgreSQL Advanced; MongoDB for Developers; Apache Airflow for Data Engineering; dbt Fundamentals.
- Blockchain Specialization — University at Buffalo & SUNY (Coursera).
- Languages: English (C2, IELTS 8.0) • Persian (Native) • Arabic (Professional) • French (Limited).